

EE 550: Optimal Control and Estimation

Course Code: EE 550

Course Name: Optimal Control and Estimation

L-T-P-C: 3-0-0-3

Prerequisites: For B.Tech- Control System or equivalent, For all- Linear Algebra or equivalent

Students intended for: M.Tech /M. Tech (R) / Ph.D. / B.Tech 3rd / 4th year

Elective or Core: Elective

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Introduction:** Review on probability and statistics, Dynamical systems, Continuous time system, Discrete time system, State space representation, their basic stability properties in time and frequency domain, Performance measures. Static optimization: Optimization with and without constraints. [5 Lectures]
- **Calculus of variations:** Fundamental concepts, Functional of a single function and several independent functions, Piecewise smooth externals, Constrained extremals. [5 Lectures]
- **Variational approach to optimal control problems:** Necessary conditions for optimal control, Algebraic Riccati equation, Linear regulator problems, Pontryagin's minimum (maximum) principle and state inequality constraints, Minimum time and control effort problems, Singular interval in optimal control problems, Two-point boundary value problem. [6 Lectures]
- **Dynamic programming:** Optimal control law, principle of optimality, interpolation, recurrence relation of dynamic programming, discrete linear quadratic regulator problem, continuous linear regulator problem, Hamilton-Jacobi-Bellman equation. [5 Lectures]
- **Output feedback and structured control:** Linear quadratic regulator with output feedback, Reference input tracking, Explicit model-following design, Output feedback in game theory and decentralized control. [6 Lectures]
- **Optimal state estimation:** Least squares estimation, Propagation of uncertainty in dynamic systems, Linear optimal state estimation (Kalman filter) for continuous and discrete time systems, Extended Kalman filters, Linear quadratic Gaussian controller, Hm filter. [10 Lectures]

Text books:

1. Robert F. Stengel, **Optimal control and estimation**, Dover Publications, 1994
2. Donald E. Kirk, **Optimal control theory: An introduction**, Dover publications, 1998

References:

1. Frank L. Lewis, Draguna Vrabie, Vassilis L. Syrmos, **Optimal control**, John Wiley & Sons, 2012
2. Dan Simon, **Optimal state estimation**, John Wiley & sons, 2006
3. Brian D.O. Anderson, John B. Moore, **Optimal control**, Prentice-Hall International, 1989.